CSR102: Design Thinking and Complex Problem Solving

The Definitive University Exam Companion & Solved Question Paper Book

Complete Assessment Suite (Sets A & B for CA-1, Mid-Term, CA-2, End-Term)

Featuring Step-by-Step Analytical Solutions & Marking Rubrics

- **Full CA-1 Papers (Units 1 & 2)**
- **Full Mid-Term Papers (Units 1, 2, 3)**
- **Full CA-2 Papers (Units 4 & 5)**
- **Full End-Term Papers (Units 1 to 6)**

Prepared for Lovely Professional University (LPU) Computer Science & Engineering

Course Outcomes CO1–CO6 • Bloom's Taxonomy L1–L5 • Standard Scoring Rubrics

LPU Engineering Academic Series

Academic Session 2026–2027

Curriculum Blueprint & Course Outcomes Mapping

Course Outcome	Unit	Competency Statement	Bloom's Level
CO1	Unit 1	Understand and apply foundations of design thinking and categorize well-structured, ill-structured, and wicked problems.	L2 Understand / L3 Apply
CO2	Unit 2	Formulate human-centered problem statements, analyze stakeholder dynamics, and conduct root cause analysis.	L3 Apply / L4 Analyze
CO3	Unit 3	Deploy analytical reasoning, evaluate cognitive biases, and apply Multi-Criteria Decision Analysis (MCDA).	L4 Analyze / L5 Evaluate
CO4	Unit 4	Generate divergent solutions using SCAMPER, Brainwriting, and apply Pugh Matrices and 2×2 prioritization.	L3 Apply / L4 Analyze
CO5	Unit 5	Design service blueprints, execute MVP validation experiments, and compute quantitative usability metrics (SUS, TCR).	L4 Analyze / L5 Evaluate
CO6	Unit 6	Structure persuasive executive presentations via Minto's SCQA and apply reflective cycles (Kolb, Gibbs, Post-mortem).	L3 Apply / L5 Evaluate

Table 1: CSR102 Course Outcomes & Bloom's Taxonomy Alignment Matrix

Examination Structure & Weightage Overview

- **Continuous Assessment 1 (CA-1):** 30 Marks, 60 Minutes. 6 Analytical Questions (5 Marks each). Covers CO1 and CO2.
- **Mid-Term Examination:** 30 Marks, 90 Minutes. Section A (10 MCQs, 10 Marks) + Section B (2 Analytical Problems, 10 Marks) + Section C (1 Case Study, 10 Marks). Covers CO1, CO2, CO3.
- **Continuous Assessment 2 (CA-2):** 30 Marks, 60 Minutes. 6 Applied Design Problems (5 Marks each). Covers CO4 and CO5.
- **End-Term Comprehensive Examination:** 50 Marks, 180 Minutes. Section A (10 MCQs, 10 Marks) + Section B (4 Scenarios, 20 Marks) + Section C (2 Capstone Challenges, 20 Marks). Covers CO1 through CO6.

Contents

1 Continuous Assessment 1 (CA-1) Suite **1**
1.1 Continuous Assessment 1 (CA-1) — Set A 2
1.2 Continuous Assessment 1 (CA-1) — Set B 5

2 Mid-Term Examination Suite (Units 1, 2 & 3) **9**
2.1 Mid-Term Examination — Set A 10
2.2 Mid-Term Examination — Set B 17

3 Continuous Assessment 2 (CA-2) Suite **25**
3.1 Continuous Assessment 2 (CA-2) — Set A 26
3.2 Continuous Assessment 2 (CA-2) — Set B 29

4 End-Term Comprehensive Examination Suite (Units 1 to 6) **34**
4.1 End-Term Comprehensive Examination — Set A 35
4.2 End-Term Comprehensive Examination — Set B 45

Chapter 1

Continuous Assessment 1 (CA-1) Suite

Continuous Assessment 1 evaluates students on **Unit 1 (Foundations of Problem Solving and Design Thinking)** and **Unit 2 (Problem Definition and Analysis)**. Each paper contains **6 analytical design questions** carrying **5 marks each** (Total = **30 Marks**, Time Allowed = **60 Minutes**).

1.1 Continuous Assessment 1 (CA-1) — Set A

LOVELY PROFESSIONAL UNIVERSITY

Transforming Education, Transforming India

Roll No.: .....
Reg. No.: .....
Section: .....

Course Code: CSR102 **Course Title:** Design Thinking and Complex Problem Solving
Academic Task Type: Continuous Assessment (CA-1) **Set:** A
Maximum Marks: 30 Marks **Time Allowed:** 60 Minutes

Course Outcomes Addressed: CO1 (Foundations & Wicked Problems), CO2 (Problem Definition & Analysis)

Note: Attempt all six questions. Each question carries 5 marks.

Q.1	Contrast well-structured (tame) problems with ill-structured wicked problems across four specific criteria defined by Rittel and Webber (1973). Provide one real-world engineering example of each.	**CO1**	L2 Understand	**5 Marks**
Q.2	Explain the non-linear relationship between the *Prototype* and *Empathize* stages in the Stanford d.school Design Thinking framework. How does testing a prototype generate fresh empathy?	**CO1**	L3 Apply	**5 Marks**
Q.3	A smart city municipal corporation observes severe resistance from taxi drivers following the rollout of an automated dispatch app. Construct a Mendelow Power-Interest Matrix classifying four distinct stakeholder groups and define the engagement strategy for each.	**CO2**	L3 Apply	**5 Marks**
Q.4	Apply the 5-Whys root cause diagnostic methodology to investigate why a university student registration portal crashed during peak course enrollment hours. Show each step of the causal chain.	**CO2**	L4 Analyze	**5 Marks**
Q.5	Construct an Ishikawa (Fishbone) diagram utilizing the 4P framework (People, Policy, Procedures, Plant/Technology) to analyze the causes of high medication dispensing errors in an outpatient pharmacy.	**CO2**	L4 Analyze	**5 Marks**
Q.6	Formulate a complete Point-of-View (POV) statement for a rural school teacher struggling to deliver hybrid digital lessons during recurring electrical blackouts. Derive three distinct "How Might We" (HMW) opportunity questions.	**CO2**	L3 Apply	**5 Marks**

Solutions and Marking Rubrics — CA-1 Set A

Step-by-Step Analytical Solution **Solution to Question 1**

1. Four Distinct Rittel & Webber Criteria:
• *Definitive Formulation:* Tame problems have clear, bounded equations (e.g., sorting an array). Wicked problems have no definitive formulation; describing the problem requires choosing an ideological stance.
• *Stopping Rule:* Tame problems terminate when an algorithm satisfies an invariant. Wicked problems terminate when resources, budget, or political patience expire.
• *Evaluation of Solutions:* Tame solutions are True/False or Correct/Incorrect. Wicked solutions are judged on a spectrum of Good/Bad or Better/Worse by competing stakeholders.
• *Reversibility:* Tame problems permit risk-free sandbox trial-and-error. Wicked problems are one-shot operations with irreversible societal consequences.
2. Engineering Examples:
• *Tame Example:* Designing a CRC32 checksum verification function in network packet parsing.
• *Wicked Example:* Designing a congestion-pricing algorithm for a major metropolitan downtown corridor.

- 4 Criteria accurately explained with theoretical grounding: **3 Marks**
- 2 Distinct, relevant engineering examples: **2 Marks**

Step-by-Step Analytical Solution **Solution to Question 2**

Non-Linear Dynamics between Prototyping and Empathy:
Traditional engineering views prototyping as a one-way progression: Concept → Build → Final Delivery. In Design Thinking, a prototype is fundamentally an **empathy instrument**.

- When an extreme user is handed a low-fidelity prototype (e.g., cardboard mock-up or paper wireframe), their natural interactions reveal unvoiced assumptions, emotional anxieties, and cognitive workarounds that passive interviews fail to surface.
- Watching a user hesitate, express confusion, or misinterpret a label provides visceral observational empathy, sending the design team directly back into the *Empathize* or *Define* stage to reframe user needs.

- Explanation of prototype as an empathy-generation tool: **3 Marks**
- Description of non-linear iterative loop (back to Empathize/Define): **2 Marks**

Step-by-Step Analytical Solution **Solution to Question 3**

Mendelow Stakeholder Classification for Taxi Dispatch System:

Stakeholder	Power	Interest	Actionable Strategy
Municipal Transport Dept	High	High	**Manage Closely:** Partner in regulatory compliance and route policy.
Taxi Fleet Owners Union	High	Low	**Keep Satisfied:** Maintain dialogue on revenue share to prevent strikes.
Individual Taxi Drivers	Low	High	**Keep Informed:** Active user training, vernacular UI, transparent fares.
Occasional Tourists	Low	Low	**Monitor:** Track app store reviews and download conversion.

- Correct 2×2 classification of 4 stakeholder groups: **2 Marks**
- Distinct, realistic engagement strategy for each quadrant: **3 Marks**

Step-by-Step Analytical Solution **Solution to Question 4**

5-Whys Diagnostic Chain for Course Enrollment Crash:
• **Why 1:** Why did the portal crash? Because backend server CPU utilization pegged at 100%, dropping HTTP requests.
• **Why 2:** Why was CPU pegged at 100%? Because 8,000 concurrent students were executing un-cached database queries simultaneously at 9:00 AM.
• **Why 3:** Why were queries un-cached? Because the session cache TTL was set to zero to avoid stale enrollment counts.
• **Why 4:** Why did all 8,000 students log in at exactly 9:00 AM? Because course registration opened simultaneously across all four academic batches.
• **Why 5 (Root Cause):** The university enrollment policy lacked batch-wise staggered registration windows and queue-throttled admission gating.

- Logical, unbroken causal chain connecting symptom to root cause: **4 Marks**
- Identification of systemic/policy failure rather than mere scapegoating: **1 Mark**

Step-by-Step Analytical Solution **Solution to Question 5**

Ishikawa 4P Breakdown of Medication Dispensing Errors:
• **People:** Pharmacist fatigue during 12-hour shifts; cognitive overload from phone calls; look-alike/sound-alike drug name confusion.
• **Policy:** Absence of mandatory double-verification rules for high-potency drugs; punitive culture discouraging near-miss reporting.
• **Procedures:** Handwritten illegible prescription routing; batch-dispensing during rush hours without individualized trays.
• **Plant / Technology:** Outdated barcode scanner latency ($\geq$ 3s delay); poor lighting in medication stockroom; cluttered inventory shelves.

- Correct categorization into all 4P dimensions: **2 Marks**
- Realistic, contextually specific failure causes under each branch: **3 Marks**

Step-by-Step Analytical Solution **Solution to Question 6**

1. POV Statement:
"A dedicated rural school teacher [User] needs reliable, asynchronous interactive teaching tools [Need] because erratic daily grid outages create severe student learning disruption and emotional helplessness [Insight]."
2. Three Derived HMW Questions:
• *HMW 1 (Leverage unexpected resource):* How might we utilize local solar battery storage or passive reflective chalkboards to maintain lesson continuity during power cuts?
• *HMW 2 (Remove the bad):* How might we eliminate the requirement for continuous real-time internet connectivity through offline peer-to-peer lesson caching?
• *HMW 3 (Amp up the good):* How might we transform sudden power cuts into interactive group problem-solving exercises that students anticipate with excitement?

- Complete POV following [User] + [Need] + [Insight] formula: **2 Marks**
- Three creative, diverse HMW questions: **3 Marks**

1.2 Continuous Assessment 1 (CA-1) — Set B

LOVELY PROFESSIONAL UNIVERSITY

Transforming Education, Transforming India

Roll No.: .....
Reg. No.: .....
Section: .....

Course Code: CSR102 **Course Title:** Design Thinking and Complex Problem Solving
Academic Task Type: Continuous Assessment (CA-1) **Set:** B
Maximum Marks: 30 Marks **Time Allowed:** 60 Minutes

Course Outcomes Addressed: CO1 (Foundations & Wicked Problems), CO2 (Problem Definition & Analysis)

Note: Attempt all six questions. Each question carries 5 marks.

Q.1	Apply the Cynefin sense-making framework to classify four distinct engineering operational domains (Clear, Complicated, Complex, Chaotic). Provide the recommended decision-making sequence for each.	**CO1**	L2 Understand	**5 Marks**
Q.2	Explain the first diamond of the UK Design Council Double Diamond model. Why does premature convergence during problem discovery lead to system failure?	**CO1**	L3 Apply	**5 Marks**
Q.3	Construct an Empathy Map (Says, Thinks, Does, Feels, Pains, Gains) for an elderly patient attempting to navigate an automated hospital appointment kiosk.	**CO2**	L3 Apply	**5 Marks**
Q.4	Differentiate between symptoms and root causes. Illustrate how an engineering team might mistake a symptom for a root cause in an e-commerce shopping cart abandonment scenario.	**CO2**	L4 Analyze	**5 Marks**
Q.5	Explain the purpose of Scope Boundary Critique. Construct an In-Scope vs. Out-of-Scope boundary table for a project deploying an AI chatbot for student academic advising.	**CO2**	L3 Apply	**5 Marks**

Q.6	Define the SMART criteria for engineering project objectives. Transform two poorly drafted objectives ("Make the app faster" and "Improve customer satisfaction") into fully compliant SMART objectives.	**CO2**	L3 Apply	**5 Marks**

Solutions and Marking Rubrics — CA-1 Set B

Step-by-Step Analytical Solution **Solution to Question 1**

Cynefin Framework Domain Breakdown:
1. **Clear (Simple):** Known causes and effects. *Response:* Sense → Categorize → Respond. Apply Best Practice (e.g., password reset procedure).
2. **Complicated:** Multiple viable solutions; requires expert diagnostic evaluation. *Response:* Sense → Analyze → Respond. Apply Good Practice (e.g., database performance index tuning).
3. **Complex:** Emergent patterns; cause and effect only visible in retrospect. *Response:* Probe → Sense → Respond. Apply Safe-to-Fail experiments and agile prototypes (e.g., redesigning social networking feed ranking).
4. **Chaotic:** High turbulence; immediate crisis. *Response:* Act → Sense → Respond. Establish immediate command and stabilization (e.g., live ransomware attack mitigation).

- Correct identification of all 4 domains: **2 Marks**
- Accurate response sequences and engineering actions: **3 Marks**

Step-by-Step Analytical Solution **Solution to Question 2**

The First Diamond (Discover & Define):
• *Discover (Divergence):* Exploring the full context, gathering empathy data from diverse users, and challenging hidden assumptions.
• *Define (Convergence):* Synthesizing observations into a clear root problem statement and design brief.
Consequences of Premature Convergence:
Premature convergence occurs when engineers accept the client's initial brief at face value without question. This leads to the "Solution in Search of a Problem" anti-pattern, where expensive software or hardware is built to solve a superficial symptom while the root human need remains completely unaddressed.

- Accurate description of Discover and Define phases: **3 Marks**
- Analysis of premature convergence risks and failure modes: **2 Marks**

Step-by-Step Analytical Solution **Solution to Question 3**

Empathy Map for Elderly Kiosk User:
• **Says:** "Where do I insert my Medicare card?", "Is this thing going to print my receipt?"
• **Does:** Squints at small touch targets; leans close to screen; hesitates before pressing buttons; clutches paper records tightly.
• **Thinks:** "I hope I don't break this machine"; "Everyone in line behind me is judging how slow I am."
• **Feels:** Visceral embarrassment; vulnerability; anxiety over losing their appointment slot.
• **Pains:** Tiny fonts, confusing medical terminology, timeout resets that erase progress, high glare on screen.
• **Gains:** Clear physical receipt with large bold room numbers, friendly assistance, dignity in self-service.

- Realistic entries across all 4 core quadrants (Says, Does, Thinks, Feels): **3 Marks**
- Deep synthesis of Pains and Gains: **2 Marks**

Step-by-Step Analytical Solution **Solution to Question 4**

Symptoms vs. Root Causes in Cart Abandonment:

• *Symptom:* Observable surface manifestation of distress (e.g., 68% of users abandon checkout on step 3).

• *Superficial Mistake:* Management assumes users abandoned carts because "the checkout button is too small" or "users lack discounts," spending $30,000 on coupon promotional pop-ups.

• *Root Cause Analysis:* Deep user shadowing reveals that step 3 demands mandatory account creation with password complexity rules and unexpected $14 shipping fees disclosed for the first time. The root cause is surprise pricing and high registration friction, not lack of interest.

- Rigorous theoretical distinction: **2 Marks**
- Detailed, realistic e-commerce failure illustration: **3 Marks**

Step-by-Step Analytical Solution **Solution to Question 5**

Scope Boundary Ledger for AI Advising Chatbot:

In-Scope (Hard Deliverables)	Out-of-Scope (Explicit Exclusions)
Automated retrieval of prerequisite courses and graduation requirements from course catalog.	Automatic registration or fee payment processing.
Natural language Q&A regarding add/drop deadlines.	Handling mental health counseling or crisis intervention (diverted to human helpline).
Integration with student portal via single sign-on (SSO).	Modifying student academic grade transcripts.

- Definition and importance of Scope Boundary Critique: **2 Marks**
- Clear, professional In-Scope vs. Out-of-Scope ledger: **3 Marks**

Step-by-Step Analytical Solution **Solution to Question 6**

SMART Objective Transformation:

Poor Objective 1: "Make the app faster."

→ **SMART Formulation:** "Reduce 95th-percentile home screen API load latency from 3.8 seconds to under 800 milliseconds across 4G mobile connections within 60 calendar days of sprint launch."

Poor Objective 2: "Improve customer satisfaction."

→ **SMART Formulation:** "Elevate post-support Net Promoter Score (NPS) from +14 to $\geq$ +45 and decrease average first-response resolution time from 24 hours to under 2 hours by the conclusion of Q3."

- Explanation of SMART dimensions: **1 Mark**
- Flawless conversion of Objective 1 and Objective 2 with exact metrics: **4 Marks**

Chapter 2

Mid-Term Examination Suite (Units 1, 2 & 3)

The Mid-Term Examination is an assessment evaluating students across **Unit 1 (Foundations)**, **Unit 2 (Problem Definition)**, and **Unit 3 (Analytical & Critical Thinking)**. Each paper consists of **Section A (Objective MCQs)**, **Section B (Applied Analytical Scenarios)**, and **Section C (Comprehensive Case Challenge)** (Total = **30 Marks**, Time Allowed = **90 Minutes**).

2.1 Mid-Term Examination — Set A

LOVELY PROFESSIONAL UNIVERSITY

Transforming Education, Transforming India

Roll No.: . . . . . **Reg. No.:** . . . . . **Section:** . . . . . .

Course Code: CSR102 **Course Title:** Design Thinking and Complex Problem Solving
Academic Task Type: Mid-Term Examination **Set:** A
Maximum Marks: 30 Marks **Time Allowed:** 90 Minutes

Course Outcomes Addressed: CO1 (Foundations), CO2 (Problem Definition), CO3 (Analytical & Critical Thinking)

Section A: Multiple Choice Questions [10 Marks]

Attempt all ten questions. Each question carries 1 mark.

Q.1 What is the primary hallmark of a wicked problem according to Rittel and Webber? [1 Mark]

(A) It can be definitively formulated
(B) It has no stopping rule
(C) It is easily solved by algorithms
(D) It has zero social impact

Q.2 In the Stanford d.school framework, which stage directly precedes Ideation? [1 Mark]

(A) Empathize
(B) Define
(C) Prototype
(D) Test

Q.3 Which quadrant of the Empathy Map captures direct quotes spoken by users during field interviews? [1 Mark]

(A) Thinks
(B) Does
(C) Says
(D) Feels

Q.4 In Mendelow's Matrix, what is the strategy for high power, low interest stakeholders? [1 Mark]

(A) Manage Closely
(B) Keep Satisfied
(C) Monitor
(D) Keep Informed

Q.5 What type of reasoning proceeds from specific empirical observations to general probabilistic principles? [1 Mark]

(A) Deductive
(B) Inductive
(C) Abductive
(D) Reductive

Q.6 Which cognitive bias involves selectively seeking data that validates pre-existing architectural beliefs? [1 Mark]

(A) Availability Bias
(B) Anchoring Effect
(C) Confirmation Bias
(D) Framing Effect

Q.7 What is the primary purpose of Barbara Minto's MECE principle? **[1 Mark]**

(A) To maximize code execution speed
(B) To eliminate overlap and ensure zero blind spots
(C) To generate rapid paper prototypes
(D) To reduce server memory usage

Q.8 In an Ishikawa diagram for services, what do the 4Ps stand for? **[1 Mark]**

(A) Plan, Price, Product, Promotion
(B) People, Policy, Procedures, Plant/Tech
(C) Past, Present, Pending, Progress
(D) Physics, Principles, Proofs, Protocols

Q.9 In an MCDA model, what must the sum of all normalized criteria weights $\sum w_j$ equal? **[1 Mark]**

(A) 0.0
(B) 1.0
(C) 10.0
(D) 100.0

Q.10 A team continues funding a failing tech stack because $500k was spent. This illustrates: **[1 Mark]**

(A) Sunk Cost Fallacy
(B) Halo Effect
(C) Overconfidence Bias
(D) Automation Bias

Section B: Applied Analytical Problems [10 Marks]

Attempt both questions. Each question carries 5 marks.

Q.11 A hospital diagnostic laboratory is evaluating three competing blood biochemistry analyzer machines using Multi-Criteria Decision Analysis (MCDA). The evaluation criteria and normalized weights are: Testing Speed ($w_1 = 0.35$), Accuracy ($w_2 = 0.30$), Capital Cost ($w_3 = 0.20$), and Vendor SLA Support ($w_4 = 0.15$). The raw performance ratings (scale 1–10) are:

- **Machine A:** Speed = 8, Accuracy = 9, Cost = 6, Support = 7
- **Machine B:** Speed = 9, Accuracy = 8, Cost = 8, Support = 6
- **Machine C:** Speed = 6, Accuracy = 10, Cost = 5, Support = 9

Calculate the composite performance score for each machine and determine the optimal selection.

CO3 L3 Apply **5 Marks**

Q.12 Deconstruct the systemic risks of introducing mandatory biometric facial recognition for campus dormitories using the MECE framework across four distinct categories.

CO3 L4 Analyze **5 Marks**

Section C: Comprehensive Design Case Scenario [10 Marks]

Attempt the comprehensive case study below.

Q.13 **Case Context:** An urban municipal library system notices a 55% decline in adolescent and young adult visits over three years, while local youth report high academic anxiety and lack of collaborative community workspace. City council proposed spending $2.5 million to purchase 300 VR gaming headsets to "attract kids back."

Deliverables Required:

(a) Analyze why purchasing VR headsets commits the "Solution in Search of a Problem" fallacy.

(b) Formulate a human-centered Point-of-View (POV) statement for urban youth.

(c) Derive three actionable "How Might We" (HMW) opportunity questions.

(d) Construct

CO2 L5 Evaluate 10 Marks

Solutions and Marking Rubrics — Mid-Term Set A

Step-by-Step Analytical Solution **Solutions to Section A: Objective MCQs**

1. (B) It has no stopping rule **2.** (B) Define **3.** (C) Says **4.** (B) Keep Satisfied **5.** (B) Inductive **6.** (C) Confirmation Bias **7.** (B) To eliminate overlap and ensure zero blind spots **8.** (B) People, Policy, Procedures, Plant/Tech **9.** (B) 1.0 **10.** (A) Sunk Cost Fallacy

1 Mark per correct MCQ = **10 Marks Total**.

Step-by-Step Analytical Solution **Solution to Question 11: MCDA Scoring Calculation**

Apply the MCDA formula $S_i = \sum_{j=1}^{4} w_j \cdot r_{ij}$:
• **Machine A Score:**
$S_A = (0.35 \times 8) + (0.30 \times 9) + (0.20 \times 6) + (0.15 \times 7) = 2.80 + 2.70 + 1.20 + 1.05 = \mathbf{7.75}$
• **Machine B Score:**
$S_B = (0.35 \times 9) + (0.30 \times 8) + (0.20 \times 8) + (0.15 \times 6) = 3.15 + 2.40 + 1.60 + 0.90 = \mathbf{8.05}$
• **Machine C Score:**
$S_C = (0.35 \times 6) + (0.30 \times 10) + (0.20 \times 5) + (0.15 \times 9) = 2.10 + 3.00 + 1.00 + 1.35 = \mathbf{7.45}$
Selection Conclusion:
Machine B achieves the highest composite performance score ($S_B = 8.05$), primarily driven by superior operational speed and balanced cost-effectiveness.

- Correct mathematical formulation and weight application: **2 Marks**
- Accurate arithmetic computation for Machines A, B, C: **2 Marks**
- Clear analytical justification and final selection: **1 Mark**

Step-by-Step Analytical Solution **Solution to Question 12: MECE Risk Deconstruction**

Four MECE Dimensions for Dormitory Facial Recognition:
1. **Technical & Reliability Risks (ME Category 1):** Lighting variations causing false rejections; network latency during peak curfew hours; sensor degradation in winter conditions.
2. **Data Privacy & Cybersecurity Risks (ME Category 2):** Centralized biometric database exposure to breach; third-party vendor data retention; lack of biometric encryption-at-rest.
3. **Regulatory & Legal Compliance Risks (ME Category 3):** Violation of personal data protection statutes; absence of explicit opt-in informed consent; liability for misidentification.
4. **Social & Inclusion Risks (ME Category 4):** Algorithmic bias and disparate error rates across demographics; student psychological feelings of constant institutional surveillance.

- Demonstrated compliance with MECE (zero overlap, exhaustive coverage): **2 Marks**
- Comprehensive, realistic risk identification under all 4 branches: **3 Marks**

Step-by-Step Analytical Solution **Solution to Question 13: Library System Design Challenge**

(a) Critique of the VR Headset Fallacy (2 Marks):
The council leaped directly to an expensive technological artifact without understanding underlying user needs. VR gaming headsets address entertainment, not collaborative study, community belonging, or academic anxiety. It commits the classic "gadget-first" engineering trap.
(b) POV Statement (2 Marks):
"Urban high school and college students [User] need a modern, psychological-safe, and collaborative co-working third place [Need] because rigid silence rules and isolated home study foster acute

academic anxiety and loneliness [Insight]."

(c) Three Derived HMW Questions (3 Marks):

• *HMW 1:* How might we zone libraries into dynamic collaborative makerspaces without alienating traditional silent readers?

• *HMW 2:* How might we empower students to co-create their own peer tutoring and social events inside the library?

• *HMW 3:* How might we integrate cafe and charging amenities to support extended hybrid group study?

(d) 5-Whys Diagnostic Chain (3 Marks):

• *Why 1:* Why do youth avoid the library? Because staff shush them when collaborating.

• *Why 2:* Why do staff shush them? Because library policy mandates strict acoustic silence throughout the building.

• *Why 3:* Why is silence mandated everywhere? Because the facility layout was built in 1965 with open unpartitioned reading halls.

• *Why 4:* Why has the layout not changed? Because past funding was allocated to books rather than modular collaborative furniture.

• *Why 5 (Root Cause):* The library operating charter treats libraries as passive book warehouses rather than active community collaboration hubs.

- Part (a) Critical analysis: **2 Marks**
- Part (b) POV statement: **2 Marks**
- Part (c) 3 HMW questions: **3 Marks**
- Part (d) 5-Whys chain: **3 Marks**

2.2 Mid-Term Examination — Set B

LOVELY PROFESSIONAL UNIVERSITY

Transforming Education, Transforming India

Roll No.: .....
Reg. No.: .....
Section: .....

Course Code: CSR102 **Course Title:** Design Thinking and Complex Problem Solving
Academic Task Type: Mid-Term Examination **Set:** B
Maximum Marks: 30 Marks **Time Allowed:** 90 Minutes

Course Outcomes Addressed: CO1 (Foundations), CO2 (Problem Definition), CO3 (Analytical & Critical Thinking)

Section A: Multiple Choice Questions [10 Marks]

Attempt all ten questions. Each question carries 1 mark.

Q.1 Which of the following problems is an example of a well-structured engineering problem? [1 Mark]

(A) Combating urban gentrification

(B) Calculating the eigenvalues of a 4×4 matrix

(C) Resolving ethical dilemmas in autonomous vehicle crashes

(D) Redesigning primary healthcare delivery

Q.2 What is the primary objective of the Double Diamond's Discover phase? [1 Mark]

(A) To rapidly converge on code

(B) To diverge and gather broad contextual insights

(C) To deploy the final commercial product

(D) To calculate unit manufacturing costs

Q.3 Which quadrant of the Empathy Map captures non-verbal behaviors and physical pathways? [1 Mark]

(A) Says
(C) Does
(B) Thinks
(D) Feels

Q.4 Which stakeholder group in Mendelow's Matrix possesses high power and high interest? [1 Mark]

(A) Monitor
(C) Keep Satisfied
(B) Keep Informed
(D) Manage Closely

Q.5 What form of logical inference is known as "Inference to the Best Explanation"? [1 Mark]

(A) Deductive Reasoning
(C) Abductive Reasoning
(B) Inductive Reasoning
(D) Syllogistic Reasoning

Q.6 What cognitive bias leads teams to overvalue options framed as avoiding losses over gaining equivalent rewards? [1 Mark]

(A) Loss Aversion / Framing Effect
(C) Anchoring Effect
(B) Availability Heuristic
(D) Dunning-Kruger Effect

Q.7 What does the acronym SMART stand for in project objective formulation? [1 Mark]

(A) Speedy, Managed, Agile, Reliable, Timely
(C) Standardized, Modular, Actionable, Robust, Technical
(B) Specific, Measurable, Achievable, Relevant, Time-bound
(D) Strategic, Monetary, Analytical, Rigorous, Trackable

Q.8 In an Ishikawa diagram, the main problem or defect is positioned at: [1 Mark]

(A) The tail of the fish
(C) The ribs of the fish
(B) The head (spine terminus) of the fish
(D) The scales of the fish

Q.9 Which sense-making domain in Cynefin involves emergent order and safe-to-fail experiments? [1 Mark]

(A) Clear
(C) Complex
(B) Complicated
(D) Chaotic

Q.10 When an engineer blindly trusts an algorithmic diagnostic output over physical symptoms, they exhibit: [1 Mark]

(A) Automation Bias
(C) Halo Effect
(B) Status Quo Bias
(D) Groupthink

Section B: Applied Analytical Problems [10 Marks]

Attempt both questions. Each question carries 5 marks.

Q.11 A university is selecting an enterprise Cloud Learning Management System (LMS) from three shortlisted vendors. The evaluation criteria and weights are: Usability ($w_1 = 0.30$), API Extensibility ($w_2 = 0.25$), Information Security Compliance ($w_3 = 0.25$), and Annual Licensing Cost ($w_4 = 0.20$). Raw ratings (scale 1–10) are:

- **Vendor Alpha:** Usability = 9, APIs = 7, Security = 8, Cost = 5
- **Vendor Beta:** Usability = 7, APIs = 9, Security = 9, Cost = 6
- **Vendor Gamma:** Usability = 6, APIs = 8, Security = 7, Cost = 9

Execute an MCDA calculation to rank all three vendors and recommend the optimal selection.

CO3 L3 Apply **5 Marks**

Q.12 Identify and analyze three distinct cognitive biases that typically distort software development estimations during sprint planning. Formulate an actionable debiasing heuristic for each.

CO3 L4 Analyze **5 Marks**

Section C: Comprehensive Design Case Scenario [10 Marks]

Attempt the comprehensive case study below.

Q.13 **Case Context:** An agricultural cooperative seeks to minimize water wastage across 1,200 family-owned orchards. Currently, farmers rely on traditional flood irrigation every Thursday, causing 40% water evaporation and soil salinization. A foreign drone vendor offered to sell thermal imaging drones to inspect orchards for $150,000.

Deliverables Required:

(a) Analyze why purchasing surveillance drones fails to address the human root problem.

(b) Construct a 5-Whys diagnostic chain explaining why farmers flood their fields on Thursdays.

(c) Formulate a human-centered POV statement for orchard farm-

Solutions and Marking Rubrics — Mid-Term Set B

Step-by-Step Analytical Solution **Solutions to Section A: Objective MCQs**

1. (B) Calculating the eigenvalues of a 4×4 matrix **2.** (B) To diverge and gather broad contextual insights **3.** (C) Does **4.** (D) Manage Closely **5.** (C) Abductive Reasoning **6.** (A) Loss Aversion / Framing Effect **7.** (B) Specific, Measurable, Achievable, Relevant, Time-bound **8.** (B) The head (spine terminus) of the fish **9.** (C) Complex **10.** (A) Automation Bias

1 Mark per correct MCQ = **10 Marks Total**.

Step-by-Step Analytical Solution **Solution to Question 11: MCDA Vendor Ranking**

Calculate $S_i = \sum_{j=1}^{4} w_j \cdot r_{ij}$:
• **Vendor Alpha:**
$S_\alpha = (0.30 \times 9) + (0.25 \times 7) + (0.25 \times 8) + (0.20 \times 5) = 2.70 + 1.75 + 2.00 + 1.00 = \mathbf{7.45}$
• **Vendor Beta:**
$S_\beta = (0.30 \times 7) + (0.25 \times 9) + (0.25 \times 9) + (0.20 \times 6) = 2.10 + 2.25 + 2.25 + 1.20 = \mathbf{7.80}$
• **Vendor Gamma:**
$S_\gamma = (0.30 \times 6) + (0.25 \times 8) + (0.25 \times 7) + (0.20 \times 9) = 1.80 + 2.00 + 1.75 + 1.80 = \mathbf{7.35}$
Recommendation:
Vendor Beta ranks first ($S_\beta = 7.80$), followed by Alpha (7.45) and Gamma (7.35). Beta provides superior API extensibility and security compliance while maintaining balanced usability.

- Accurate calculation for all three vendors: **3 Marks**
- Complete ranking and justified recommendation: **2 Marks**

Step-by-Step Analytical Solution **Solution to Question 12: Cognitive Biases in Sprint Planning**

1. **Overconfidence / Planning Fallacy:**
• *Manifestation:* Developers consistently underestimate task completion time by 40–50%, assuming zero interruptions or bugs.
• *Debiasing Heuristic:* Reference-Class Forecasting: base sprint estimates on historical velocity data from similar past tasks rather than optimistic mental projections.
2. **Anchoring Effect:**
• *Manifestation:* The team lead mentions "this looks like a 3-point story," anchoring the entire room around 3 points.
• *Debiasing Heuristic:* Planning Poker: all developers reveal story point cards simultaneously in secret, preventing anchor contamination.
3. **Groupthink / Social Desirability:**
• *Manifestation:* Junior engineers withhold concerns regarding legacy dependencies to avoid conflict with senior architects.
• *Debiasing Heuristic:* Anonymous pre-sprint risk submission and required devil's advocate reviews.

- Accurate analysis of 3 biases in sprint context: **3 Marks**
- Concrete, realistic debiasing interventions: **2 Marks**

Step-by-Step Analytical Solution **Solution to Question 13: Orchard Irrigation Case Study**

(a) Critique of Drone Surveillance Fallacy (2 Marks):
Thermal imaging drones provide aerial data on soil temperature but do not automate water control, cannot change canal availability schedules, and cost more than the annual cooperative budget. It creates high technical overhead without resolving the physical and behavioral bottlenecks.

(b) 5-Whys Diagnostic Chain (3 Marks):
• *Why 1:* Why do farmers flood their fields on Thursday? Because canal water is released into secondary ditches only on Thursdays.
• *Why 2:* Why is canal water released only on Thursdays? Because the irrigation department uses a fixed weekly rotation schedule.
• *Why 3:* Why do farmers apply excessive water? Because they fear canal water might be delayed next week, so they over-saturate soil as insurance.
• *Why 4:* Why do they lack real-time soil moisture knowledge? Because physical laboratory testing is slow and expensive.
• *Why 5 (Root Cause):* The system lacks on-site visual soil moisture feedback and flexible on-demand micro-irrigation scheduling.

(c) POV Statement (2 Marks):
"Family orchard farmers [User] need simple, trusted real-time soil moisture indicators and controlled water delivery [Need] because rigid canal schedules and fear of drought compel them to flood fields destructively [Insight]."

(d) Three Derived HMW Questions (3 Marks):
• *HMW 1:* How might we provide low-cost mechanical color-changing soil moisture probes requiring zero batteries?
• *HMW 2:* How might we organize cooperative night-drip irrigation scheduling to eliminate daytime evaporation?
• *HMW 3:* How might we incentivize water conservation through shared pumping credits?

- Part (a) Critical failure analysis: **2 Marks**
- Part (b) 5-Whys diagnostic chain: **3 Marks**
- Part (c) POV statement: **2 Marks**
- Part (d) 3 actionable HMW questions: **3 Marks**

Chapter 3

Continuous Assessment 2 (CA-2) Suite

Continuous Assessment 2 evaluates students on **Unit 4 (Ideation and Creative Problem Solving)** and **Unit 5 (Workflow Design, Testing, and Iterative Refinement)**. Each paper contains **6 applied engineering design challenges** carrying **5 marks each** (Total = **30 Marks**, Time Allowed = **60 Minutes**).

3.1 Continuous Assessment 2 (CA-2) — Set A

LOVELY PROFESSIONAL UNIVERSITY

Transforming Education, Transforming India

Roll No.: .....
Reg. No.: .....
Section: .....

Course Code: CSR102 **Course Title:** Design Thinking and Complex Problem Solving
Academic Task Type: Continuous Assessment (CA-2) **Set:** A
Maximum Marks: 30 Marks **Time Allowed:** 60 Minutes

Course Outcomes Addressed: CO4 (Ideation & Selection), CO5 (Workflow Design & Validation)

Note: Attempt all six questions. Each question carries 5 marks.

Q.1	Apply the SCAMPER creative ideation methodology to generate at least six innovative system concepts to redesign the physical campus library experience.	**CO4**	L3 Apply	**5 Marks**
Q.2	Contrast traditional open-room brainstorming with the Brainwriting 6-3-5 protocol. Explain why Brainwriting produces higher conceptual variety and minimizes psychological bias.	**CO4**	L2 Understand	**5 Marks**
Q.3	Explain the difference between a Wizard of Oz prototype and a Concierge MVP. Provide an illustrative software application example for each.	**CO5**	L2 Understand	**5 Marks**
Q.4	Construct a Multi-Tier Service Blueprint for an online grocery home delivery platform. Diagram the Line of Interaction, Line of Visibility, and identify at least two Failure Nodes [!].	**CO5**	L4 Analyze	**5 Marks**
Q.5	In a usability test of a new mobile banking interface across 10 participants, 8 participants successfully transferred funds without assistance, taking an average of 42 seconds. Calculate the Task Completion Rate (TCR) and evaluate the outcome against industry benchmarks.	**CO5**	L3 Apply	**5 Marks**
Q.6	A usability evaluation yields the following 10 raw System Usability Scale (SUS) scores from a participant: $[5, 1, 4, 2, 5, 1, 4, 2, 5, 1]$. Calculate the overall composite SUS score and determine its usability grade.	**CO5**	L3 Apply	**5 Marks**

Solutions and Marking Rubrics — CA-1 Set A

Step-by-Step Analytical Solution **Solution to Question 1: SCAMPER Application**

SCAMPER Redesign for Campus Library:
• **Substitute (S):** Substitute static printed textbooks with scannable dynamic AR holograms and digital annotations.
• **Combine (C):** Combine silent study cubicles with sound-isolated podcast and video recording studios.
• **Adapt (A):** Adapt the Amazon fulfillment center logistics model by installing automated retrieval robots for archival books.
• **Modify / Magnify (M):** Magnify operating hours to 24/7/365 with automated biometric security pods.
• **Put to Other Use (P):** Repurpose underutilized basement book stacks into collaborative student incubation makerspaces.
• **Eliminate (E):** Eliminate manual checkout desks through RFID doorway arch sensors that auto-charge loans to student IDs.
• **Reverse / Rearrange (R):** Invert book acquisition: allow students to vote on library procurement budgets directly.

- Six distinct SCAMPER categories correctly addressed: **3 Marks**
- Quality, feasibility, and originality of proposed concepts: **2 Marks**

Step-by-Step Analytical Solution **Solution to Question 2: Brainwriting 6-3-5 vs. Brainstorming**

Pathologies of Traditional Brainstorming:
1. *Production Blocking:* Only one participant can speak at a time; others forget or abandon ideas while waiting.
2. *Evaluation Apprehension:* Junior members fear looking foolish in front of senior leaders.
3. *Social Loafing:* Disengaged participants hide behind talkative dominant individuals.

The 6-3-5 Mechanism and Psychological Advantages:
• 6 participants $\times$ 3 ideas $\times$ 6 rounds $=$ 108 ideas in 30 minutes.
• Total silence eliminates vocal domination and interruptions.
• Round-robin passing forces systematic cross-pollination and evolutionary idea building without social hierarchy friction.

- Identification of 3 brainstorming pathologies: **2 Marks**
- Explanation of 6-3-5 structure and psychological advantages: **3 Marks**

Step-by-Step Analytical Solution **Solution to Question 3: Wizard of Oz vs. Concierge MVP**

Dimension	Wizard of Oz Prototype	Concierge MVP
User Perception	Front-end appears fully automated (AI/software-driven).	Service is explicitly disclosed as 100% manual human white-glove support.
Backstage Reality	Humans secretly execute backend tasks manually behind the scenes.	Humans openly assist the customer directly in person or via call.
Validation Goal	Tests interface UX, trust, user journey, and perceived responsiveness.	Tests deep customer willingness to pay and customized service workflow.
Example	An automated legal contract redline app where human lawyers manually edit text within 10 minutes.	A personal shopping stylist who accompanies the user to clothing stores.

- Clear structural comparison across perception and backend: **3 Marks**
- Realistic, distinct software/business examples for both: **2 Marks**

Step-by-Step Analytical Solution **Solution to Question 4: Grocery Platform Service Blueprint**

Service Blueprint Architectural Layers:
• **Physical Evidence:** Mobile app UI catalog; order status tracker; SMS delivery pin; grocery paper bag.
• **Customer Actions:** Browses produce → adds to cart → selects delivery time slot → completes payment → receives groceries.
— **Line of Interaction** —
• **Frontstage Contact:** Delivery driver arrives at doorstep, hands over bag, confirms delivery pin.
— **Line of Visibility** —
• **Backstage Contact:** Warehouse picker gathers items with barcode scanner; packer bags items; dispatch manager routes driver.
— **Line of Internal Interaction** —
• **Support Processes:** Real-time warehouse inventory sync; credit card gateway settlement; automated route optimization algorithm.
• **Failure Nodes [!]:**
1. *[! 1] Out-of-Stock Discrepancy:* In-app produce shows available, but warehouse bin is empty during picking.
2. *[! 2] Payment Timeout Gateway Failure:* Bank server drops confirmation during session token handover.

- Inclusion of all key blueprint tiers with lines of interaction/visibility: **3 Marks**
- Identification and analysis of 2 critical failure points: **2 Marks**

Step-by-Step Analytical Solution **Solution to Question 5: Task Completion Rate (TCR) Calculation**

Formula:

$$\text{TCR} = \frac{\text{Successful Task Completions}}{\text{Total Task Attempts}} \times 100\% = \frac{8}{10} \times 100\% = \mathbf{80.0}\%$$

Benchmarking & Evaluation:
• In consumer mobile financial applications, an 80% TCR represents an acceptable baseline, but

falls below the 90% threshold required for mission-critical core tasks (such as funds transfer).
• The 2 failed attempts (20% drop-off) signal critical friction, likely caused by confusing beneficiary account number entry or OTP verification timeouts.

- Correct formula and mathematical calculation: **2 Marks**
- Professional benchmarking evaluation and failure diagnosis: **3 Marks**

Step-by-Step Analytical Solution **Solution to Question 6: System Usability Scale (SUS) Score**

Given Responses: $R = [5, 1, 4, 2, 5, 1, 4, 2, 5, 1]$
Odd questions (1, 3, 5, 7, 9) contribute $(R_i - 1)$:
• $(5-1) + (4-1) + (5-1) + (4-1) + (5-1) = 4+3+4+3+4 = 18$
Even questions (2, 4, 6, 8, 10) contribute $(5 - R_i)$:
• $(5-1) + (5-2) + (5-1) + (5-2) + (5-1) = 4+3+4+3+4 = 18$
Composite Calculation:

$$\text{SUS Score} = 2.5 \times (18 + 18) = 2.5 \times 36 = \mathbf{90.0}$$

Usability Grade Interpretation:
A score of 90.0 places the system in the top 5% of all benchmarked digital products (Grade A+, "Best Imaginable" usability), far exceeding the industry average of 68.0.

- Correct formula application for odd and even items: **3 Marks**
- Accurate calculation resulting in 90.0: **1 Mark**
- Accurate interpretation against industry percentile benchmarks: **1 Mark**

3.2 Continuous Assessment 2 (CA-2) — Set B

LOVELY PROFESSIONAL UNIVERSITY

Transforming Education, Transforming India

Roll No.: .....
Reg. No.: .....
Section: .....

Course Code: CSR102 **Course Title:** Design Thinking and Complex Problem Solving
Academic Task Type: Continuous Assessment (CA-2) **Set:** B
Maximum Marks: 30 Marks **Time Allowed:** 60 Minutes

Course Outcomes Addressed: CO4 (Ideation & Selection), CO5 (Workflow Design & Validation)

Note: Attempt all six questions. Each question carries 5 marks.

Q.1	Explain the three lenses of innovation according to IDEO: Desirability, Feasibility, and Viability (DFV). Illustrate how a solution that satisfies any two lenses while ignoring the third inevitably fails.	**CO4**	L2 Understand	**5 Marks**
Q.2	Construct a Pugh Controlled Convergence Matrix evaluating two candidate smart parking mobile app concepts against the existing municipal street parking baseline.	**CO4**	L3 Apply	**5 Marks**

Q.3	Categorize four engineering backlog features for an electric vehicle charging station network using a 2×2 Impact vs. Effort Prioritization Matrix.	**CO4**	L3 Apply	**5 Marks**
Q.4	Design a Smoke Test (Fake Door experiment) to validate customer demand for an automated drone laundry pickup service in high-rise condominium complexes.	**CO5**	L4 Analyze	**5 Marks**
Q.5	Explain the architecture of the 4-Quadrant Feedback Capture Grid (Likes, Criticisms, Questions, Ideas) and demonstrate how it is used to structure post-usability sprint backlogs.	**CO5**	L3 Apply	**5 Marks**
Q.6	A usability evaluation yields the following 10 raw SUS scores: $[3, 3, 3, 3, 3, 3, 3, 3, 3, 3]$. Calculate the composite SUS score, explain its mathematical meaning, and evaluate the result.	**CO5**	L3 Apply	**5 Marks**

Solutions and Marking Rubrics — CA-2 Set B

Step-by-Step Analytical Solution **Solution to Question 1: IDEO DFV Triad**

The Three Lenses of Innovation:
1. **Desirability (Human):** Do people actually want it? Does it resolve a genuine pain point?
2. **Feasibility (Technical):** Can it be built with available engineering, software, and physical constraints?
3. **Viability (Business):** Can it generate sustainable operational revenue or societal return on investment?

Failure Modes of Two-Lens Traps:
• *Desirable + Feasible, but Not Viable:* Free unlimited cloud storage. Users love it, engineers can build it, but storage costs bankrupt the enterprise.
• *Desirable + Viable, but Not Feasible:* Teleportation for commuters. Massive consumer demand and immense commercial profit, but physically impossible.
• *Feasible + Viable, but Not Desirable:* Technically robust enterprise time-tracking dongle that employees hate and actively bypass.

- Accurate explanation of all 3 lenses: **2 Marks**
- Detailed failure mode analysis with realistic illustrations: **3 Marks**

Step-by-Step Analytical Solution **Solution to Question 2: Pugh Matrix for Smart Parking**

Pugh Decision Matrix (Baseline Datum = Physical Coin Parking Meter):

Criteria	Baseline (0)	Concept A: Sensor App	Concept B: Community Valet
Driver Search Time	0	+1	+1
Payment Friction	0	+1	0
Hardware Maintenance Cost	0	-1	+1
Data Privacy Protection	0	0	-1
Scalability	0	+1	-1
Net Score ($\Sigma+ - \Sigma-$)	**0**	**+2 Net**	**0 Net**

Conclusion: Concept A (IoT Sensor Parking App) achieves +2 net score, outperforming the baseline and valet model due to superior digital payment convenience and enterprise scalability.

- Correct Pugh Matrix structure with baseline datum: **2 Marks**
- Logical scoring across all criteria and clear net synthesis: **3 Marks**

Step-by-Step Analytical Solution **Solution to Question 3: 2×2 Impact vs. Effort Prioritization**

EV Charging Station Feature Prioritization:
1. **Quick Wins (High Impact, Low Effort):** Real-time charger availability status on

Google/Apple Maps via lightweight API webhook.
2. **Strategic Bets / Major Projects (High Impact, High Effort):** Automated robotic underbody plug-in charging mechanism for hands-free docking.
3. **Fill-ins (Low Impact, Low Effort):** Custom ambient LED lighting color selection during charging.
4. **Money Pits (Low Impact, High Effort):** Built-in touch screen video streaming games in the outdoor charger cabinet (vulnerable to rain, high vandalism, low demand).

- Correct allocation across all 4 quadrants: **2 Marks**
- Highly realistic engineering features with clear operational rationale: **3 Marks**

Step-by-Step Analytical Solution **Solution to Question 4: Smoke Test Experiment Design**

Drone Laundry Smoke Test Architecture:
1. **Artifact:** A high-fidelity responsive landing page targeted via localized social media ads to residents of a 40-story residential tower, advertising "Automated 1-Hour Rooftop Drone Dry-Cleaning Pickup".
2. **The Call to Action (CTA):** A prominent button stating "Select Laundry Package & Schedule Pickup".
3. **Friction Hurdle / Commitment Signal:** When clicked, users are prompted to enter their apartment number and credit card for a $5 reservation deposit.
4. **The Reveal / Ethical De-escalation:** Upon entering data, a dialogue appears: "We are currently calibrating flight paths for your tower. Your card has not been charged. You are #4 on the priority alpha list!"
5. **Validation Metric:** If $> 12\%$ of visitors submit the reservation form, demand is validated; otherwise, pivot concept.

- End-to-end experiment design from traffic to conversion: **3 Marks**
- Ethical disclosure and quantitative validation metric definition: **2 Marks**

Step-by-Step Analytical Solution **Solution to Question 5: Feedback Capture Grid**

Feedback Capture Grid Structure:
• **Top Left ($\oplus$ Likes):** What users praised unprompted (e.g., "One-click reorder is amazing"). Action: Preserve and elevate in UI.
• **Top Right ($\triangle$ Criticisms):** Confusion, errors, and drop-offs (e.g., "Where do I find my invoice?"). Action: Create immediate P0 bug tickets.
• **Bottom Left (? Questions):** Ambiguities raised during usage (e.g., "Is my data encrypted?"). Action: Add contextual help tooltips.
• **Bottom Right ($\star$ Ideas):** Spontaneous user suggestions (e.g., "Can you auto-split the bill with roommates?"). Action: Add to product backlog for sprint refinement.

- Complete definition of all 4 quadrants: **2 Marks**
- Direct operational linkage to sprint backlog prioritization: **3 Marks**

Step-by-Step Analytical Solution **Solution to Question 6: Neutral SUS Score Calculation**

Given Responses: All 10 items scored 3 (Neutral).
Odd questions (1, 3, 5, 7, 9): $(3 - 1) = 2 \times 5 = 10$
Even questions (2, 4, 6, 8, 10): $(5 - 3) = 2 \times 5 = 10$

$$\text{SUS Score} = 2.5 \times (10 + 10) = 2.5 \times 20 = \mathbf{50.0}$$

Significance and Interpretation:

• An exact neutral response on all 10 Likert items produces a score of exactly **50.0**.

• A score of 50.0 falls well below the industry benchmark of 68.0, earning a grade of "Poor / High Friction". Neutrality in usability signifies lack of delight, moderate confusion, and high churn risk.

- Flawless calculation resulting in 50.0: **2 Marks**
- Mathematical and behavioral interpretation of the neutral baseline: **3 Marks**

Chapter 4

End-Term Comprehensive Examination Suite (Units 1 to 6)

The End-Term Final Examination is the comprehensive capstone assessment spanning **all 6 Units** of the CSR102 curriculum. Each paper contains **Section A (10 Objective MCQs, 10 Marks)**, **Section B (4 Analytical Scenarios, 20 Marks)**, and **Section C (2 Full-Scale Design Challenges, 20 Marks)** (Total = **50 Marks**, Time Allowed = **180 Minutes / 3 Hours**).

4.1 End-Term Comprehensive Examination — Set A

LOVELY PROFESSIONAL UNIVERSITY

Transforming Education, Transforming India

Roll No.: . . . . . **Reg. No.:** . . . . . **Section:** . . . . .

Course Code: CSR102 **Course Title:** Design Thinking and Complex Problem Solving
Academic Task Type: End-Term Comprehensive Examination **Set:** A
Maximum Marks: 50 Marks **Time Allowed:** 180 Minutes (3 Hours)

Course Outcomes Addressed: CO1–CO6 (Comprehensive Syllabus Units 1 to 6)

Section A: Objective Multiple Choice Questions [10 Marks]

Attempt all ten questions. Each question carries 1 mark.

Q.1 Which problem solving approach is best suited for complex, non-linear human problems? [1 Mark]

(A) Design Thinking
(B) Linear Waterfall Method
(C) Trial-and-Error without reflection
(D) Premature Optimization

Q.2 In the Stanford d.school framework, what is the primary deliverable of the Define stage? [1 Mark]

(A) A functional codebase
(B) Point-of-View (POV) statement
(C) An invoice for services
(D) A usability survey

Q.3 Which tool maps user emotions across touchpoints over time? [1 Mark]

(A) Fishbone Diagram
(B) Customer Journey Map
(C) Pareto Chart
(D) Gantt Chart

Q.4 What does the 'A' in the SCAMPER creative ideation methodology stand for? [1 Mark]

(A) Analyze
(B) Adapt
(C) Automate
(D) Accelerate

Q.5 In Brainwriting 6-3-5, how many total ideas are produced after 6 rounds? [1 Mark]

(A) 36
(B) 72
(C) 108
(D) 216

Q.6 What is the primary characteristic of a Wizard of Oz prototype? [1 Mark]

(A) It uses expensive robotics
(B) The front-end appears automated while back-end tasks are done manually
(C) It only exists as code without UI
(D) It is tested without real users

Q.7 An industry average System Usability Scale (SUS) score is approximately: **[1 Mark]**

(A) 30.0
(C) 68.0
(B) 50.0
(D) 95.0

Q.8 In Barbara Minto's SCQA presentation framework, what does 'C' stand for? **[1 Mark]**

(A) Conclusion
(C) Context
(B) Complication
(D) Criteria

Q.9 What is the first stage in David Kolb's Experiential Learning Cycle? **[1 Mark]**

(A) Concrete Experience
(C) Abstract Conceptualization
(B) Reflective Observation
(D) Active Experimentation

Q.10 In a blameless postmortem, errors are primarily attributed to: **[1 Mark]**

(A) Careless junior employees
(C) Malicious intent
(B) Systemic and architectural flaws
(D) Bad luck

Section B: Applied Analytical Engineering Scenarios [20 Marks]

Attempt all four questions. Each question carries 5 marks.

Q.11 A smart mobility startup is selecting an electric battery chemistry for electric fleet delivery scooters. The criteria and weights are: Energy Density ($w_1 = 0.35$), Thermal Safety ($w_2 = 0.30$), Lifecycle Degradation ($w_3 = 0.20$), and Sourcing Cost ($w_4 = 0.15$). The ratings (1–10) are:

- **Chemistry A (NMC):** Density = 9, Safety = 6, Lifecycle = 7, Cost = 6
- **Chemistry B (LFP):** Density = 7, Safety = 9, Lifecycle = 9, Cost = 8

Execute an MCDA calculation to determine the optimal battery chemistry for the commercial fleet.

CO3 L3 Apply **5 Marks**

Q.12 Deconstruct the Root Causes of high student dropout rates in online asynchronous computer programming courses using an Ishikawa (Fishbone) diagram across the 4Ps (People, Policy, Procedures, Plant/Technology).

CO2 L4 Analyze **5 Marks**

Q.13 Construct a Multi-Tier Service Blueprint for a self-service electric vehicle battery swapping station. Include Customer Actions, Frontstage, Backstage, Support Processes, and at least two Failure Nodes [!].

CO5 L4 Analyze **5 Marks**

Q.14 Structure an Executive Pitch Presentation using Barbara Minto's Pyramid Principle (SCQA) proposing to replace passwords with FIDO2 hardware passkeys in an enterprise bank.

CO6 L3 Apply **5 Marks**

Section C: Comprehensive Capstone Design Challenges [20 Marks]

Attempt both comprehensive design challenges. Each question carries 10 marks.

Q.15 Design Challenge 1: Urban Public Healthcare Access

An overburdened regional public health department experiences severe outpatient clinic congestion, long queues, and widespread medication shortages. Elderly patients often travel three hours by bus only to discover that prescribed hypertension medications are out of stock.

Deliverables Required:

(a) Analyze why this

Q.16 Design Challenge 2: Post-Incident Reflective Audit & Validation Architecture

A fintech peer-to-peer payments app launched a major redesign that caused a 40% drop in transaction volume and an influx of 15,000 angry customer support tickets complaining that the "Send Money" button had disappeared into an obscure nested menu.

Deliverables Required:

(a) Conduct an incident reflective

Solutions and Marking Rubrics — End-Term Set A

Step-by-Step Analytical Solution **Solutions to Section A: Objective MCQs**

1. (A) Design Thinking **2.** (B) Point-of-View (POV) statement **3.** (B) Customer Journey Map **4.** (B) Adapt **5.** (C) 108 **6.** (B) Front-end appears automated while backend tasks are done manually **7.** (C) 68.0 **8.** (B) Complication **9.** (A) Concrete Experience **10.** (B) Systemic and architectural flaws

1 Mark per correct MCQ = **10 Marks Total**.

Step-by-Step Analytical Solution **Solution to Question 11: Battery Chemistry MCDA**

Calculate $S_i = \sum w_j \cdot r_{ij}$:
• **Chemistry A (NMC):**
$S_A = (0.35 \times 9) + (0.30 \times 6) + (0.20 \times 7) + (0.15 \times 6) = 3.15 + 1.80 + 1.40 + 0.90 = \mathbf{7.25}$
• **Chemistry B (LFP):**
$S_B = (0.35 \times 7) + (0.30 \times 9) + (0.20 \times 9) + (0.15 \times 8) = 2.45 + 2.70 + 1.80 + 1.20 = \mathbf{8.15}$
Selection Conclusion:
Chemistry B (LFP) decisively outperforms Chemistry A (8.15 vs 7.25), because commercial delivery fleets prioritize thermal safety, long cycle life, and low unit costs over pure energy density.

- Accurate calculation for NMC and LFP: **3 Marks**
- Clear analytical justification and final selection: **2 Marks**

Step-by-Step Analytical Solution **Solution to Question 12: Ishikawa 4P for Coding Course Drop-out**

• **People:** Imposter syndrome among beginners; cognitive overload from syntax; lack of peer belonging; unresponsive teaching assistants.
• **Policy:** Rigid non-negotiable weekly assignment deadlines; all-or-nothing grading; zero grace periods for working adults.
• **Procedures:** Passive 90-minute lecture videos without active coding checkpoints; lack of automated scaffolding hints.
• **Plant / Technology:** Clunky web IDE environment that fails to save code states; confusing error compiler messages; lack of dark mode or mobile editor support.

- Inclusion of all 4P categories: **2 Marks**
- Deep, realistic pedagogical and technical insights: **3 Marks**

Step-by-Step Analytical Solution **Solution to Question 13: Battery Swapping Service Blueprint**

• **Physical Evidence:** Kiosk touchscreen, automated mechanical compartment door, status LED, invoice notification.
• **Customer Actions:** Pulls scooter into bay → scans QR code → removes depleted battery → inserts into slot → retrieves fresh battery → clicks into scooter.
— **Line of Interaction** —
• **Frontstage Contact:** Automated voice guidance from kiosk; mechanical locking mechanism releasing charged battery.
— **Line of Visibility** —
• **Backstage Contact:** Internal conveyor belts moving battery to high-voltage cooling rack; robotic state-of-health (SoH) diagnostics.
— **Line of Internal Interaction** —

• **Support Processes:** Cloud billing API deduction; grid load balancer managing off-peak thermal charging.
• **Failure Nodes [!]:**
1. *[! 1] Lock Jam:* Battery compartment door fails to unlatch due to thermal expansion.
2. *[! 2] Swapped Battery Degradation:* Cell with degraded capacity ($\leq 60\%$ SoH) inadvertently released to rider.

- Complete blueprint architecture across interaction and visibility lines: **3 Marks**
- Identification and mitigation of 2 critical failure nodes: **2 Marks**

Step-by-Step Analytical Solution **Solution to Question 14: Minto SCQA Passkeys Executive Pitch**

• **Situation (S):** Enterprise bank serves 4 million active mobile users handling \$2.8B in annual transactions using SMS OTP and complex passwords.
• **Complication (C):** Phishing attacks and SIM-swapping fraud increased by 140% this year, inflicting \$18M in customer fraud reimbursements and generating severe reputational damage.
• **Question (Q):** How can we eliminate credential phishing and reduce login friction without increasing customer drop-off?
• **Answer (A):** Deploy biometric FIDO2 hardware passkeys natively integrated with FaceID/fingerprint biometrics, reducing login latency by 70% and mathematically eliminating credential interception.

- Complete SCQA formulation with business context: **3 Marks**
- Persuasive executive alignment connecting security to user experience: **2 Marks**

Step-by-Step Analytical Solution **Solution to Question 15: Public Healthcare Capstone Challenge**

(a) Wicked Problem Analysis (3 Marks):
1. *No definitive formulation:* The challenge intertwines medical supply chains, municipal bus schedules, rural poverty, and pharmaceutical inventory policies.
2. *Irreversible consequences:* Stock-outs cause stroke, cardiac arrest, and death; interventions cannot be tested trivially in sandboxes.
3. *Competing stakeholders:* Doctors, bus operators, pharmaceutical distributors, and patients hold conflicting objectives.
(b) POV Statement & HMW Questions (3 Marks):
• *POV:* "Elderly hypertensive patients [User] need guaranteed medication availability and decentralized local dispensing [Need] because exhausting transit journeys ending in empty pharmacy shelves cause life-threatening health crises and loss of trust [Insight]."
• *HMW 1:* How might we communicate real-time pharmacy medicine stock via basic SMS before patients leave home?
• *HMW 2:* How might we utilize existing rural post offices as secure monthly medication pick-up points?
• *HMW 3:* How might we deliver 90-day supply batches to chronic patients to reduce travel frequency by 66%?
(c) SCAMPER Ideation Concepts (4 Marks):
• **Substitute (S):** Substitute in-person visits with automated vernacular phone calls for routine prescription refills.
• **Combine (C):** Combine public postal delivery vans with pharmaceutical cold-chain parcel drops.
• **Adapt (A):** Adapt the Amazon hub locker model by placing smart medication lockers at rural bus junctions.

• **Eliminate (E):** Eliminate paper prescription renewals by auto-authorizing chronic hypertension refills upon verified blood pressure readings.

- Part (a) Wicked problem framing: **3 Marks**
- Part (b) POV statement and 3 HMW questions: **3 Marks**
- Part (c) 4 SCAMPER interventions: **4 Marks**

Step-by-Step Analytical Solution **Solution to Question 16: Gibbs Cycle Audit & Usability Sprint**

(a) Gibbs' Reflective Cycle Deployment (5 Marks):
1. **Description:** App redesign launched; transaction volume crashed 40%; 15k tickets received complaining that 'Send Money' was buried.
2. **Feelings:** Team felt defensive, arguing that the redesign was 'aesthetically cleaner'; then panic set in as revenue dropped.
3. **Evaluation:** Aesthetics improved, but core usability and discoverability were catastrophic failures.
4. **Analysis:** The team fell victim to *Designer Vanity* and confirmation bias, testing the UI only with internal designers who already knew where buttons were hidden. Zero think-aloud usability testing was conducted with authentic users.
5. **Conclusion:** Clean aesthetics must never compromise task affordances. Core transactional triggers must remain primary on home screens.
6. **Action Plan:** Immediately rollback navigation; implement mandatory usability gating (SUS ≥ 80) before any production release.

(b) 14-Day Validation Sprint Architecture (5 Marks):
• *Days 1–3:* Rapid wireframe refactor restoring prominent 'Send Money' action button on bottom navigation dock.
• *Days 4–7:* Conduct moderated usability testing with 15 representative users, tracking Task Completion Rate (TCR) and computing composite System Usability Scale (SUS) scores.
• *Days 8–10:* Populate Feedback Capture Grid: synthesize remaining Likes, Criticisms, and Questions.
• *Days 11–14:* Verify that SUS ≥ 82.0 and TCR $= 100\%$; deploy phased 10% canary release to production.

- Part (a) All 6 stages of Gibbs Cycle rigorously articulated: **5 Marks**
- Part (b) 14-day sprint protocol with SUS metrics and feedback grid: **5 Marks**

4.2 End-Term Comprehensive Examination — Set B

LOVELY PROFESSIONAL UNIVERSITY

Transforming Education, Transforming India

Roll No.: .....
Reg. No.: .....
Section: .....

Course Code: CSR102 **Course Title:** Design Thinking and Complex Problem Solving
Academic Task Type: End-Term Comprehensive Examination **Set:** B
Maximum Marks: 50 Marks **Time Allowed:** 180 Minutes (3 Hours)

Course Outcomes Addressed: CO1–CO6 (Comprehensive Syllabus Units 1 to 6)

Section A: Objective Multiple Choice Questions [10 Marks]

Attempt all ten questions. Each question carries 1 mark.

Q.1 Which framework categorizes decision contexts into Clear, Complicated, Complex, and Chaotic? [1 Mark]

(A) Double Diamond
(B) Cynefin Framework
(C) MCDA Matrix
(D) Ishikawa Model

Q.2 What is the primary formula structure for a human-centered Point-of-View (POV) statement? [1 Mark]

(A) If-Then-Else
User needs [Need] because [Surprising Insight]
(B) Input → Process → Output
(C) Problem = Symptom + Cost

Q.3 Which diagnostic tool explores causal chains by repeatedly asking 'Why'? [1 Mark]

(A) Fishbone 6M
(B) The 5-Whys Method
(C) Pugh Matrix
(D) Affinity Diagram

Q.4 What does the 'E' in SCAMPER stand for? [1 Mark]

(A) Execute
(B) Evaluate
(C) Eliminate
(D) Expand

Q.5 In Stuart Pugh's Controlled Convergence Matrix, how is the baseline datum scored across all criteria? [1 Mark]

(A) Always +1
(B) Always 0
(C) Always -1
(D) Always 10

Q.6 What does a System Usability Scale (SUS) score of 50.0 represent? [1 Mark]

(A) Top 5% usability
(B) Exact neutral response across all items
(C) Perfect system usability
(D) Zero usability

Q.7 What is the boundary between customer-visible actions and backstage operations in a Service Blueprint? [1 Mark]

(A) Line of Interaction
(B) Line of Visibility
(C) Line of Internal Coordination
(D) Line of Separation

Q.8 What are the four components of Barbara Minto's SCQA framework? [1 Mark]

(A) Speed, Cost, Quality, Action
(B) Situation, Complication, Question, Answer
(C) Scope, Criteria, Quantification, Assessment
(D) Strategy, Concept, Query, Architecture

Q.9 Which stage of Graham Gibbs' Reflective Cycle asks 'What else could you have done?' [1 Mark]

(A) Description
(B) Analysis
(C) Conclusion
(D) Feelings

Q.10 What is the primary principle of blameless engineering postmortems? **[1 Mark]**

(A) Assigning individual fault quickly

(B) Treating human error as a symptom of deeper system design flaws

(C) Issuing monetary fines

(D) Ignoring system crashes

Section B: Applied Analytical Engineering Scenarios [20 Marks]

Attempt all four questions. Each question carries 5 marks.

Q.11	A university data center is selecting an enterprise storage architecture from three alternatives: SAN, NAS, and Distributed Object Storage. The criteria and weights are: Read/Write Throughput ($w_1 = 0.30$), Scalability ($w_2 = 0.30$), Disaster Recovery Redundancy ($w_3 = 0.25$), and Total Cost of Ownership ($w_4 = 0.15$). The performance ratings (1–10) are: • **Option 1 (SAN):** Throughput = 10, Scalability = 6, Redundancy = 8, Cost = 5 • **Option 2 (NAS):** Throughput = 7, Scalability = 8, Redundancy = 7, Cost = 8 • **Option 3 (Distributed Object):** Throughput = 8, Scalability = 10, Redundancy = 9, Cost = 7 Compute the MCDA composite scores and identify the winning architecture.	**CO3**	L3 Apply	**5 Marks**
Q.12	Explain the difference between Inductive and Abductive reasoning in the context of user research and product feature discovery.	**CO3**	L2 Understand	**5 Marks**
Q.13	Design an MVP validation experiment for an on-demand drone organ transport system between regional hospitals. Choose between a Smoke Test, Wizard of Oz, or Concierge MVP and justify your selection.	**CO5**	L4 Analyze	**5 Marks**
Q.14	Construct a 2×2 Impact vs. Effort Prioritization Matrix classifying four potential software features for an automated student examination proctoring tool.	**CO4**	L3 Apply	**5 Marks**

Section C: Comprehensive Capstone Design Challenges [20 Marks]

Attempt both comprehensive design challenges. Each question carries 10 marks.

Q.15 Design Challenge 1: Smart Campus Energy Conservation

A university campus with 35,000 students experiences massive electricity waste: lecture halls have air conditioning and high-intensity lights running continuously in empty buildings, wasting $850,000 annually. The administration considered installing manual timer switches, but faculty and staff complained they

Q.16 Design Challenge 2: Service Blueprinting & Blameless Postmortem

An automated campus drone meal delivery service launched during midterms crashed spectacularly: 45 drones collided with trees, 200 orders were delivered cold or dropped onto car roofs, and students flooded social media with mocking videos.

Deliverables Required:

(a) Design a complete Multi-Tier Service Blueprint

Solutions and Marking Rubrics — End-Term Set B

Step-by-Step Analytical Solution **Solutions to Section A: Objective MCQs**

1. (B) Cynefin Framework **2.** (B) [User] needs [Need] because [Surprising Insight] **3.** (B) The 5-Whys Method **4.** (C) Eliminate **5.** (B) Always 0 **6.** (B) Exact neutral response across all items **7.** (B) Line of Visibility **8.** (B) Situation, Complication, Question, Answer **9.** (C) Conclusion **10.** (B) Treating human error as a symptom of deeper system design flaws

1 Mark per correct MCQ = **10 Marks Total**.

Step-by-Step Analytical Solution **Solution to Question 11: Data Center Storage MCDA**

Calculate $S_i = \sum w_j \cdot r_{ij}$:
• **Option 1 (SAN):**
$S_1 = (0.30 \times 10) + (0.30 \times 6) + (0.25 \times 8) + (0.15 \times 5) = 3.00 + 1.80 + 2.00 + 0.75 =$ **7.55**
• **Option 2 (NAS):**
$S_2 = (0.30 \times 7) + (0.30 \times 8) + (0.25 \times 7) + (0.15 \times 8) = 2.10 + 2.40 + 1.75 + 1.20 =$ **7.45**
• **Option 3 (Distributed Object):**
$S_3 = (0.30 \times 8) + (0.30 \times 10) + (0.25 \times 9) + (0.15 \times 7) = 2.40 + 3.00 + 2.25 + 1.05 =$ **8.70**
Recommendation:
Distributed Object Storage emerges as the definitive winner ($S_3 = 8.70$), outperforming SAN and NAS due to near-infinite scalability, robust built-in erasure coding replication, and low commodity storage costs.

- Accurate calculation across all three candidates: **3 Marks**
- Clear analytical justification and final selection: **2 Marks**

Step-by-Step Analytical Solution **Solution to Question 12: Inductive vs. Abductive in Design**

• **Inductive Reasoning (Pattern Extraction):**
Observes repeated behavioral occurrences to infer general statistical truths. Example: User telemetry shows that 74% of mobile app users drop off when asked to type their address manually $\Longrightarrow$ Inductive conclusion: Manual forms induce friction.
• **Abductive Reasoning (Inference to Best Explanation):**
Observes an anomalous, surprising behavior and makes an empathetic creative leap to hypothesize the latent root cause. Example: An older user pauses for 25 seconds on a 'Submit' button despite zero errors displayed. An inductive approach sees only delay. An abductive leap realizes: "The user fears that clicking will irrevocably charge their credit card without a confirmation step." This hypothesis directly generates a confirmation modal prototype.

- Clear epistemological distinction: **2 Marks**
- Specific, highly insightful design/user research illustrations: **3 Marks**

Step-by-Step Analytical Solution **Solution to Question 13: Drone Organ Transport MVP Selection**

Optimal Strategy: High-Fidelity Prototype Spike with Human-in-the-Loop Shadowing.
• *Why Smoke Test / Fake Door is Inappropriate:* High ethical stakes and patient life dependency make deceptive mock landing pages unethical and legally prohibited in organ logistics.
• *Validation Architecture:* Deploy a single certified drone on a pre-cleared 4-kilometer corridor

between two partner teaching hospitals to transport non-biological dummy payload bags with thermal sensors and GPS loggers.
• *Critical Hypotheses Tested:* Vibration levels, temperature maintenance (±0.5°C), landing precision in wind gusts, and handover latency between flight technician and surgical nurse.

- Rigorous critique of why simple smoke tests fail in high-risk contexts: **2 Marks**
- Detailed, actionable, ethical validation experiment protocol: **3 Marks**

Step-by-Step Analytical Solution **Solution to Question 14: Proctoring Tool 2×2 Prioritization**

Classification of Proctoring Features:
1. **Quick Wins (High Impact, Low Effort):** Instant multi-tab browser blur detection alert.
2. **Strategic Bets (High Impact, High Effort):** Low-bandwidth edge AI gaze-tracking algorithm that runs on client CPU without sending raw video feeds to the cloud.
3. **Fill-ins (Low Impact, Low Effort):** Customizable dark mode for the exam interface.
4. **Money Pits (Low Impact, High Effort):** Continuous biometric 3D facial mesh reconstruction via webcams (prohibitive client CPU load, high false rejections, severe student privacy backlash).

- Correct placement across all 4 quadrants: **2 Marks**
- Clear technical and pedagogical justification: **3 Marks**

Step-by-Step Analytical Solution **Solution to Question 15: Campus Energy Conservation Capstone**

(a) Root Cause Analysis (3 Marks):
• *5-Whys:* Why are empty rooms cooled? Because AC runs on an unmonitored master building schedule → Why master schedule? Because individual room occupancy is not tracked → Why not tracked? Because room sensors were deemed too costly in 2010 → Why do faculty not switch them off? Because switches are locked in electrical closets to prevent tampering → *Root Cause:* Centralized rigid scheduling decoupled from real-time class registration schedules.
• *Ishikawa Breakdown:* People (faculty lack awareness), Policy (facilities department locks master panels), Technology (no dynamic room booking API integration).
(b) POV Statement & HMW Questions (3 Marks):
• *POV:* "University facilities managers [User] need automated, occupant-aware ambient climate control [Need] because rigid centralized schedules and lack of room-level feedback waste massive energy and budget while frustrating occupants [Insight]."
• *HMW 1:* How might we link classroom AC power directly to the live registrar course schedule database?
• *HMW 2:* How might we empower students to 'check into' a room to activate cooling for study sessions?
• *HMW 3:* How might we gamify energy savings between campus departments through transparent floor dashboards?
(c) SCAMPER Solutions (4 Marks):
• **Substitute:** Substitute mechanical timers with calendar-integrated smart IoT thermostat relays.
• **Combine:** Combine Wi-Fi access point device count telemetry with HVAC airflow throttling.
• **Modify:** Modify cooling set-points: drift temperature dynamically by 1.5°C during 15-minute passing periods.
• **Eliminate:** Eliminate manual master electrical closet overrides entirely.

- Part (a) 5-Whys and Ishikawa analysis: **3 Marks**
- Part (b) POV statement and 3 HMW questions: **3 Marks**
- Part (c) 4 SCAMPER solutions: **4 Marks**

Step-by-Step Analytical Solution **Solution to Question 16: Drone Crash Blueprint & Blameless Postmortem**

(a) Service Blueprint Design (5 Marks):
• *Physical Evidence:* Mobile meal tracking app; insulated drone tether box; thermal bag; notification ping.
• *Customer Actions:* Orders burger → walks to outdoor campus drop circle → confirms arrival pin → unlatches box.
• *Line of Interaction:*
• *Frontstage Contact:* Drone descends to 3-meter hover, lowers tethered box into painted yellow circle.
• *Line of Visibility:*
• *Backstage Contact:* Restaurant kitchen packs hot meal into aerodynamically balanced container; ground crew snaps container onto drone hook.
• *Line of Internal Interaction:*
• *Support Processes:* Autonomous flight collision avoidance mesh; LIDAR wind-gust compensator; campus airspace traffic scheduler.
• *Failure Nodes [!]:* [! 1] High wind gusts destabilizing tether winch; [! 2] Tree canopy obscuring visual landing fiducial marker.
(b) Blameless Postmortem Protocol (5 Marks):
1. *Incident Timeline:* 12:15 PM: 200 orders surge simultaneously; 12:22 PM: GPS signal reflection off campus glass tower causes 45 drones to drift into trees; 12:35 PM: Emergency manual override server crashes under load.
2. *Systemic Contributing Factors:* Gaps in multi-path GPS filtering near glass buildings; absence of automated winch cut-off on tether snag; failure to stress-test concurrent peak meal volume in simulation.
3. *Corrective Remediation Plan:* P0 Ticket #401: Install RTK ground base stations for centimeter-accurate positioning (Due in 7 days); P0 Ticket #402: Implement failsafe auto-abort climb when tether tension exceeds 3 kg (Due in 5 days); P1 Ticket #403: Cap concurrent delivery slots at 25 drones per 15-minute window.

- Part (a) Multi-tier blueprint with failure nodes: **5 Marks**
- Part (b) Blameless postmortem with timeline, root causes, and P0/P1 remediation tickets: **5 Marks**